

Day 10: Advanced Data Modeling Techniques

📌 Definition to Remember

Advanced data modeling addresses complex relational patterns including **Role-Playing Dimensions** (multiple date keys), **Inactive Relationships** (activated via `USERELATIONSHIP`), and **Bridge Tables** for Many-to-Many resolution.

1. Role-Playing Dimensions & Inactive Relationships

- **Problem:** `Fact_Sales` has both `OrderDate`, `ShipDate`, and `DeliveryDate`, all needing the `Dim_Date` table.
- **Solution 1 (Recommended):** Create one Active relationship (`OrderDate` to `Date`) and multiple **Inactive Relationships** (dotted lines) for `ShipDate` and `DeliveryDate`.
- **Activating via DAX:**

```
Shipped Sales =  
CALCULATE(  
    [Total Sales],  
    USERELATIONSHIP(Fact_Sales[ShipDateKey], Dim_Date[DateKey])  
)
```

2. Resolving Many-to-Many (:) Relationships via Bridge Tables

- Instead of using native direct Many-to-Many relationships, introduce an intermediate distinct **Bridge Table** with 1:* relationships on both sides:

```
[Table_Students] (1) —> (*) [Bridge_Enrollment] (*) <— (1) [Table_Courses]
```

3. Parent-Child Hierarchies (Organization Charts)

- Flattening manager-employee hierarchies using DAX `PATH()` and `PATHITEM()` functions.

★ Must-Master Skills & Interview Points

1. When to use `USERELATIONSHIP()` with inactive relationships vs duplicating the Date dimension table.
2. How Bridge tables prevent Cartesian product inflation in Many-to-Many scenarios.
3. Best practice for handling multiple transaction dates in fact tables.

⚡ Quick Recall

Active Relationship (Default) + Inactive Relationship (Activated via `USERELATIONSHIP`) + Bridge Table
(*:*)

